

Problema 2 Lab 4

$$A = \{xy \mid y \in \{0,1\}^*\}$$

$$S = 0^p 1 0^p$$

$$p = 4$$

$$S = 0000100001$$

Caso 1

$$\begin{array}{c} 0000100001 \\ \underbrace{\quad\quad\quad}_x \underbrace{\quad\quad}_y \underbrace{\quad\quad}_z \end{array}$$

$$xy^2z = 00 \ 0000 \ 100001$$

$$000000 \neq 100001$$

$$|y| = 2 > 0 \quad \checkmark$$

$$|xy| = 4 \leq 4 \quad \checkmark$$

→ Pero $xy^2z \notin A$ ✗

Caso 2

$$\begin{array}{c} 0 \ 0 \ 00100001 \\ \underbrace{\quad}_x \underbrace{\quad}_y \underbrace{\quad\quad\quad}_z \end{array}$$

$$xy^2z = 0 \ 00 \ 00100001$$

$$000001 \neq 00001$$

$$|y| = 1 > 0 \quad \checkmark$$

$$|xy| = 2 \leq 4 \quad \checkmark$$

→ $xy^2z \notin A$ ✗

Caso 3

$$\begin{array}{c} 0 \ 000 \ 100001 \\ \underbrace{\quad}_x \underbrace{\quad\quad}_y \underbrace{\quad\quad\quad}_z \end{array}$$

$$xy^2z = 0 \ 000000 \ 100001$$

$$0000000 \neq 100001$$

$$|y| = 3 > 0 \quad \checkmark$$

$$|xy| = 4 \leq 4 \quad \checkmark$$

$xy^2z \notin A$ ✗